

Vault-Door-5

📁 picoCTF > Reverse Engineering · + Tags

The code give is this:

```

21 //
22 // -Minion #2414
23 public String base64Encode(byte[] input) {
24     return Base64.getEncoder().encodeToString(input);
25 }
26
27 // URL encoding is meant for web pages, so any double agent spies who steal
28 // our source code will think this is a web site or something, definitely not
29 // vault door! Oh wait, should I have not said that in a source code
30 // comment?
31 //
32 // -Minion #2415
33 public String urlEncode(byte[] input) {
34     StringBuffer buf = new StringBuffer();
35     for (int i=0; i<input.length; i++) {
36         buf.append(String.format("%%%2x", input[i]));
37     }
38     return buf.toString();
39 }
40
41 public boolean checkPassword(String password) {
42     String urlEncoded = urlEncode(password.getBytes());
43     String base64Encoded = base64Encode(urlEncoded.getBytes());
44     String expected = "JTYzJTMwJTZlJTc2JTMzJTcyJTc0JTMxJTZlJTY3JTVm"
45         + "JTY2JTYyJTMwJTZkJTVmJTYyJTYxJTM1JTY1JTVmJTM2"
46         + "JTM0JTVmJTY1JTY0JTMwJTYyJTM0JTM4JTM4";
47     return base64Encoded.equals(expected);
48 }

```

Here, it is visible that the password is first encrypted using url encodign and then, encrypted using base64 encoding. To reveal the flag, decode base64 first and url decode the given result at last.

First Decryption

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

Input

JTYzJTMwJ TZ1JTc2JTMzJTcyJ Tc0JTMxJ TZ1JTY3JTVm

Output

%63%30%6e%76%33%72%74%31%6e%67%5f

Decode from URL-encoded format

Simply enter your data then push the decode button.

%63%30%6e%76%33%72%74%31%6e%67%5f

i For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☐ Decode recursively (up to 16 times; useful when the data is encoded multiple times).

☒ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

c0nv3rt1ng_

First part of the flag: c0nv3rt1ng_

Second Decryption

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

Input

JTY2JTcyJTMwJTZkJTVmJTYyJTYxJTM1JTY1JTVmJTM2

Output

%66%72%30%6d%5f%62%61%35%65%5f%36

Decode from URL-encoded format

Simply enter your data then push the decode button.

%66%72%30%6d%5f%62%61%35%65%5f%36

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< DECODE > Decodes your data into the area below.

fr0m_ba5e_6

Second part of the flag: fr0m_ba5e_6

Third Decryption

Recipe

From Base64

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

Input
JTM0JTVmJTY1JTY0JTMwJTYyJTM0JTMMyJTM4JTM4

Output
%34%5f%65%64%30%62%34%32%38%38

Decode from URL-encoded format

Simply enter your data then push the decode button.

%34%5f%65%64%30%62%34%32%38%38

For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8

Source character set.

☐

Decode each line separately (useful for when you have multiple entries).

☐

Decode recursively (up to 16 times; useful when the data is encoded multiple times).

☒ Live mode OFF

Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE >

Decodes your data into the area below.

4_ed0b4288

Third part of the flag: 4_ed0b4288

The flag: picoCTF{c0nv3rt1ng_fr0m_ba5e_64_ed0b4288}